

Sujal Shrestha

shresthasujal2021@gmail.com | +977-9813590600 | Boudha, Kathmandu | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

BCA 7th-semester student from Kathmandu who began with a strong Java/Spring Boot backend focus and has since grown into full-stack development. Recognising the direction of Nepal's IT industry, I am actively expanding into Python, React, and AI integration — while keeping my Java foundation solid. My goal is to become a full-stack developer who can build complete products end-to-end and work with AI-powered features. I learn fast, take full ownership, and am seeking an internship — frontend, backend, full stack, or any IT role — to gain real hands-on experience, contribute to an actual team, and understand how the industry works in Nepal.

PROJECTS

Online Medical Laboratory Management System

[GitHub →](#)

Full Stack | Java • Spring Boot • Spring MVC • MySQL • Leaflet / OpenStreetMap *BCA Capstone Project*

- A full-stack, role-based web platform built entirely from scratch connecting patients with medical labs across three panels:
 - **Patients:** discover nearby labs on an interactive map, book tests, pay online, track appointment status, and download reports.
 - **Lab Staff:** manage test offerings, handle appointments, verify payments, and upload result files.
 - **Admin:** verify lab registrations and oversee all platform data.
- **Technical highlights:** Haversine formula (HaversineUtil) for real-distance nearest-lab discovery; Leaflet + OpenStreetMap for interactive map location picking during registration; BCrypt password hashing for secure auth; role-based session authorization across all panels; DB-level triggers for appointment slot conflict prevention and data integrity.

Ecommerce StoreFront

[GitHub →](#)

Full Stack | Java 17 • Spring Boot 3 • React 18 • Vite • MySQL • Spring Data JPA

- Full-stack e-commerce platform with 40 seeded products across 6 categories (Electronics, Clothing, Books, Home & Kitchen, Sports, Beauty).
- Features live search, category filtering via dropdown, and low-to-high / high-to-low price sorting — all without page reloads.
 - **Backend:** Spring Boot REST API (/api/products, /api/categories) with JPA/Hibernate and auto-seeded database on first run.
 - **Frontend:** React 18 + Vite for near-instant HMR; dark mode support; responsive layout with CSS custom properties.

Real-Time Chat Application

[GitHub →](#)

Backend + Frontend | Spring Boot 3.2.5 • WebSocket • STOMP • SockJS • Java 21

- Group chat web app with live message delivery, color-coded user avatars, animated message bubbles, and read receipts.
 - **Architecture:** Browser → SockJS → /ws endpoint → STOMP → ChatController → broadcasts to all /topic/messages subscribers.
 - **Resilience:** SockJS fallback for environments without native WebSocket; auto-reconnect with 3-second retry on disconnect.

YoutubeTools

[GitHub →](#)

Backend | Java 21 • Spring Boot • Thymeleaf • Tailwind CSS • YouTube Data API v3

- Three tools in one app: SEO tag generator (searches similar videos, extracts & copies tags), thumbnail extractor (highest-quality download), and full video metadata retriever.
 - **API integration:** YouTube Data API v3 with structured DTOs (SearchApiResponse, VideoApiResponse, Snippet, Thumbnails) for clean JSON deserialization.

CryptoTracker

[GitHub →](#)

Backend | Java 25 • Spring Boot 4 • Thymeleaf • Tailwind CSS • CoinGecko API

- Real-time cryptocurrency price tracker — live USD prices for multiple coins simultaneously with 24-hour change indicators (up/down).
- Consumes CoinGecko's free public API (no key required); JSON parsing via org.json; quick-pick buttons for popular coins.

EDUCATION

Patan Multiple Campus, Lalitpur, Nepal

Bachelor in Computer Application (BCA) — 7th Semester | Expected Graduation: 2026 Kathmandu, Nepal

SKILLS

- **Backend:** Java, Spring Boot, Spring MVC, Spring Data JPA, Servlet, JSP, REST API design, WebSocket / STOMP
- **Frontend:** HTML, CSS, Bootstrap, Tailwind CSS, JavaScript, React 18 + Vite (actively building), Thymeleaf
- **Languages:** Python (OOP), PHP, C#
- **Database & Tools:** MySQL, Git & GitHub, IntelliJ IDEA, VS Code, Apache Tomcat, XAMPP, Maven, Lombok
- **Expanding into:** Python (Django / FastAPI), React advanced patterns, AI / LLM API integration
- **Soft Skills:** Fast learner, strong ownership mindset, team collaboration, creative problem solving

INTERESTS

- Full Stack Development • Backend Architecture • AI & LLM Integration • Open Source • Nepal IT Industry